

Finance Club Open Project Summer 2025

Credit Card Behaviour Score Prediction Using Classification and Risk-Based Techniques

Akanksha Sahni | 22322002 | akanksha_s@hs.iitr.ac.in

Objective

The primary objective of this project is to develop a robust and financially interpretable binary classification model that predicts whether a credit card customer will default on their payment in the following month. By leveraging anonymized historical behavioural data of over 30,000 credit card customers—including variables such as payment history, bill amounts, repayment patterns, and demographic features—the project aims to accurately identify potential defaulters in advance. This predictive capability will enable the bank to enhance its credit risk management framework by proactively adjusting credit exposure, triggering early warning systems, and prioritizing risk-based interventions. Beyond accurate prediction, the project also seeks to generate actionable financial insights by analyzing the behavioural factors that drive default risk, engineering meaningful features, and evaluating multiple classification algorithms. Ultimately, the goal is to deliver a model that not only flags high-risk customers but also supports business decision-making through clear financial interpretation and alignment with the bank's risk appetite.

Overview & Modeling Strategy

The project follows a structured data science workflow to predict credit card defaults. The approach begins with thorough data cleaning and preprocessing, including handling missing values and ensuring correct data types. Exploratory and financial analyses are conducted to understand how key behavioral variables—like payment delays, repayment consistency, and credit utilization—impact default risk.

Feature engineering is performed to create financially meaningful variables such as credit utilization ratio and delinquency streaks. To address class imbalance in the dataset, techniques like SMOTE, class weighting, or downsampling are used.

Multiple classification models are tested and compared, including Logistic Regression, Decision Trees, and ensemble methods like XGBoost and LightGBM. Model evaluation prioritizes metrics such as F1-score, recall, and AUC-ROC, with the classification threshold set according to the bank's risk appetite. The final, tuned model is used to generate production-style predictions on the validation dataset, ensuring results are both accurate and interpretable for business decision-making.

EDA Findings and Visualizations

A comprehensive exploratory data analysis (EDA) is conducted to understand the structure, distributions, and relationships within the credit card default dataset. Below are the detailed findings, along with recommended visualizations.

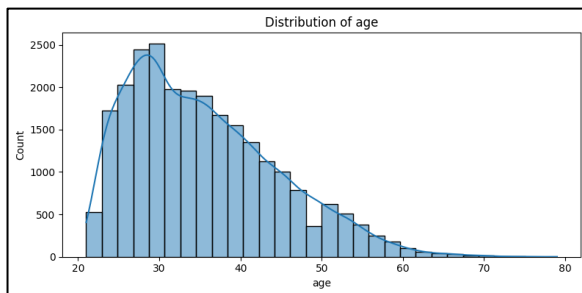
1. Data Quality and Structure

- The training dataset contains 25,247 records and 27 variables, while the validation dataset has 5,016 records and 26 variables.
- Missing values were minimal, with only the 'age' variable showing 126 missing entries in the training set; all other fields were complete.

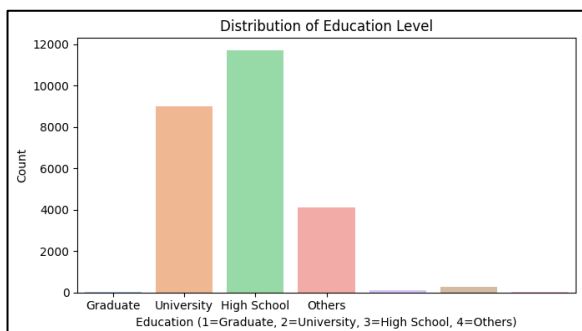
2. Univariate Analysis

➤ Demographic Variables:

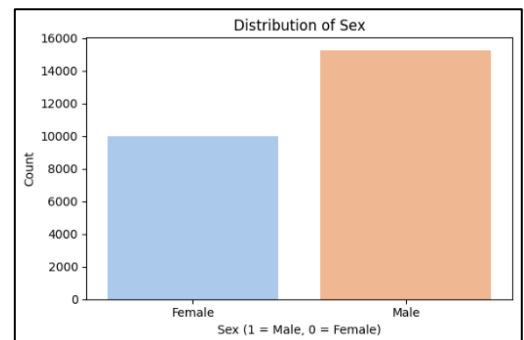
Age: The distribution of age is slightly right-skewed, with most customers between 25 and 45 years old.



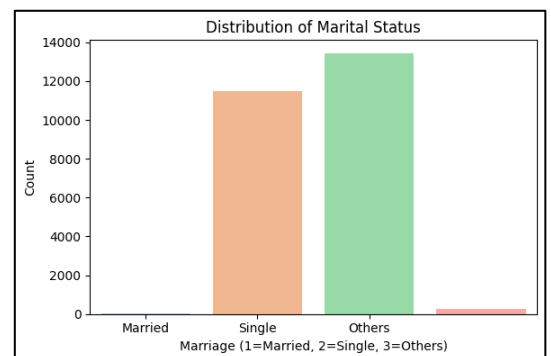
Education: Most customers have university or graduate-level education.



Sex: The dataset is relatively balanced between male and female customers.



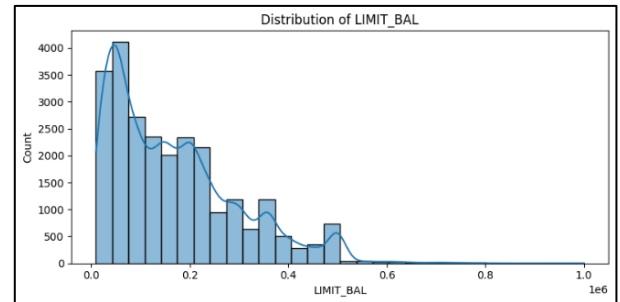
Marriage: The majority are married or single, with 'others' being a small fraction.



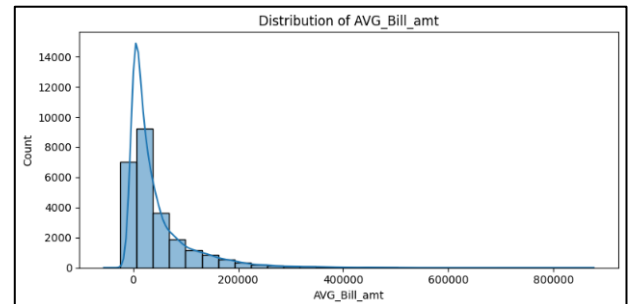
➤ **Financial Variables:**

LIMIT_BAL (Credit Limit):

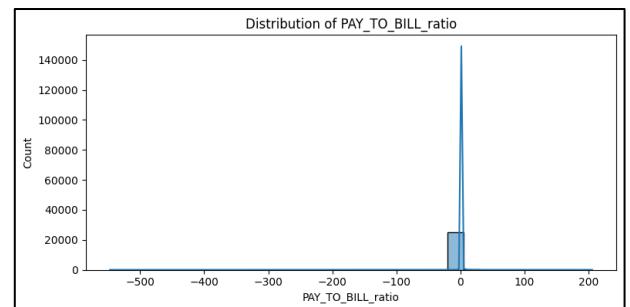
The distribution is right-skewed, with most customers having lower to moderate credit limits, and a few with very high limits.



AVG_Bill_amt: Shows a wide range, with most customers having moderate average bill amounts and a long tail of higher values.

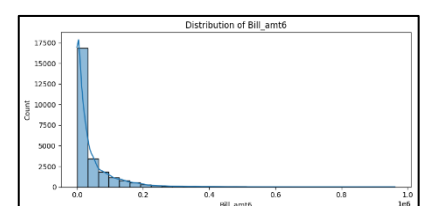
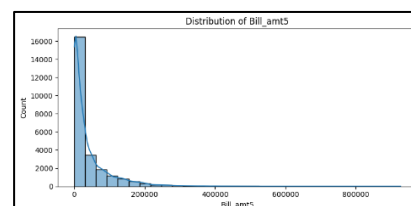
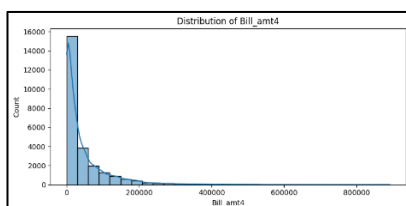
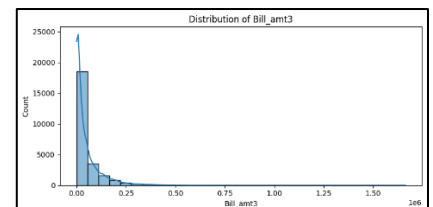
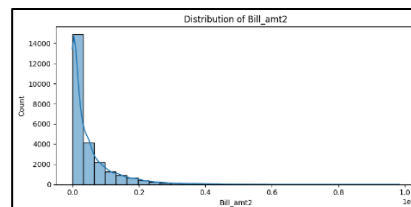
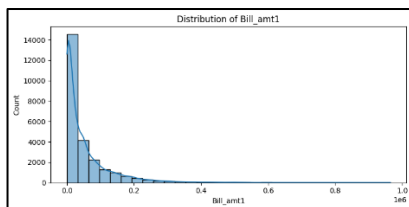


PAY_TO_BILL_ratio: Most values cluster below 1, indicating that many customers pay less than their total bill, with a minority overpaying.

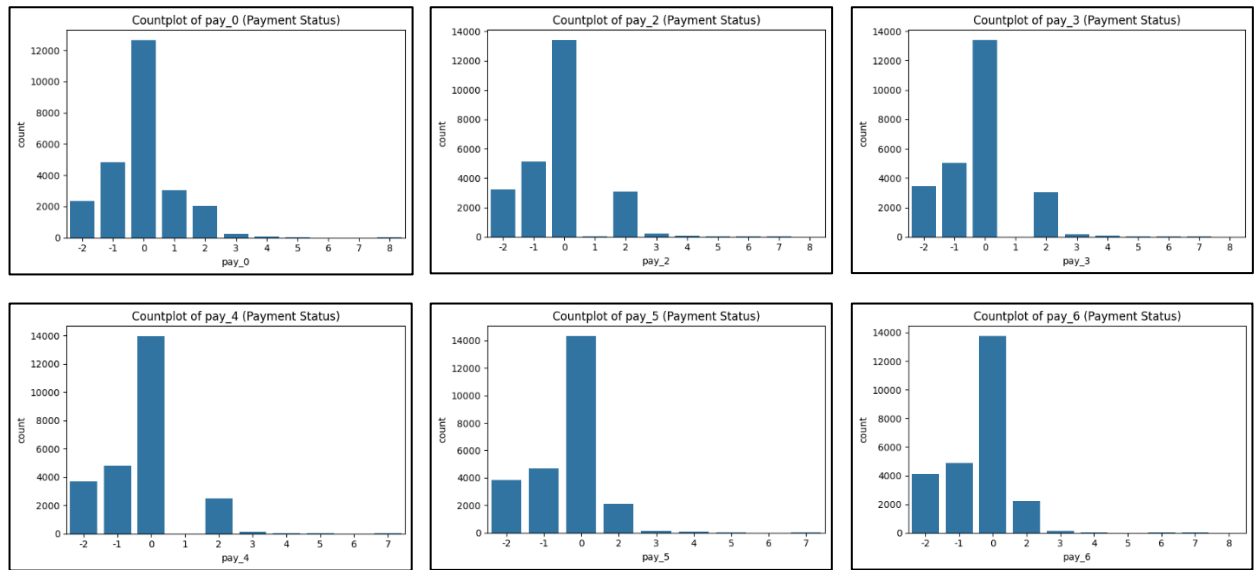


➤ **Behavioral Variables:**

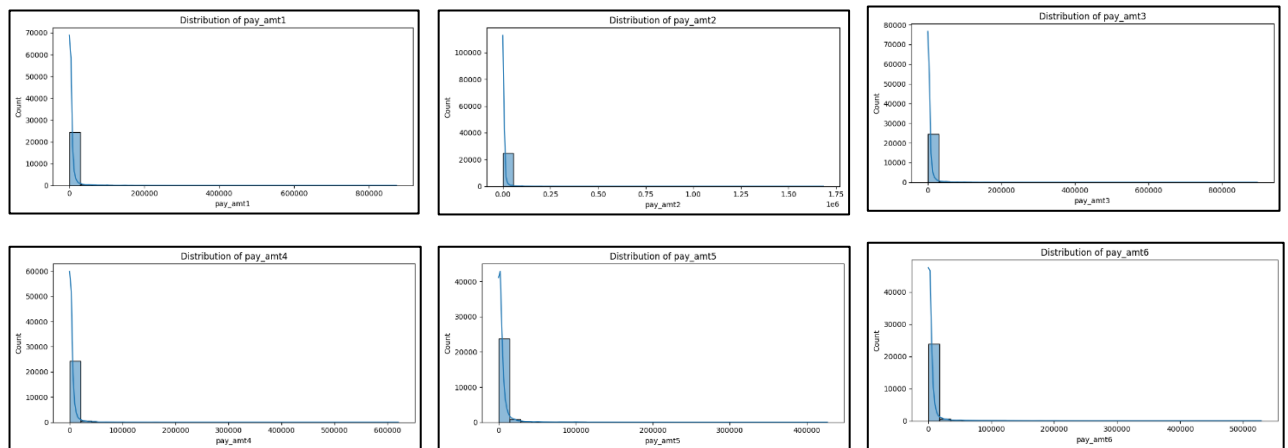
Bill_amt1 to Bill_amt6: Bill amounts are right-skewed, with most customers having moderate bills and a few with very large amounts.



PAY_0 to PAY_6 (Repayment Status): Most values are -1 or 0, indicating on-time or minimum payments, but there is a notable presence of positive values (delays), which are more common among defaulters.

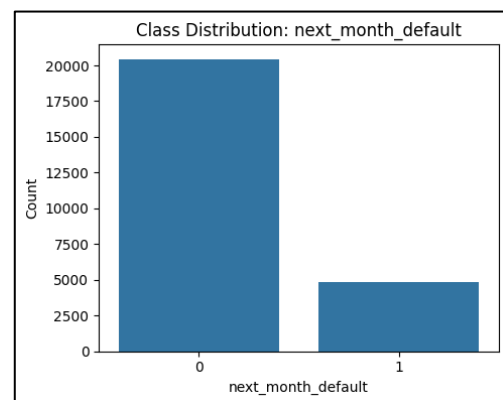


pay_amt1 to pay_amt6: Payment amounts are also right-skewed, with most customers making smaller payments.



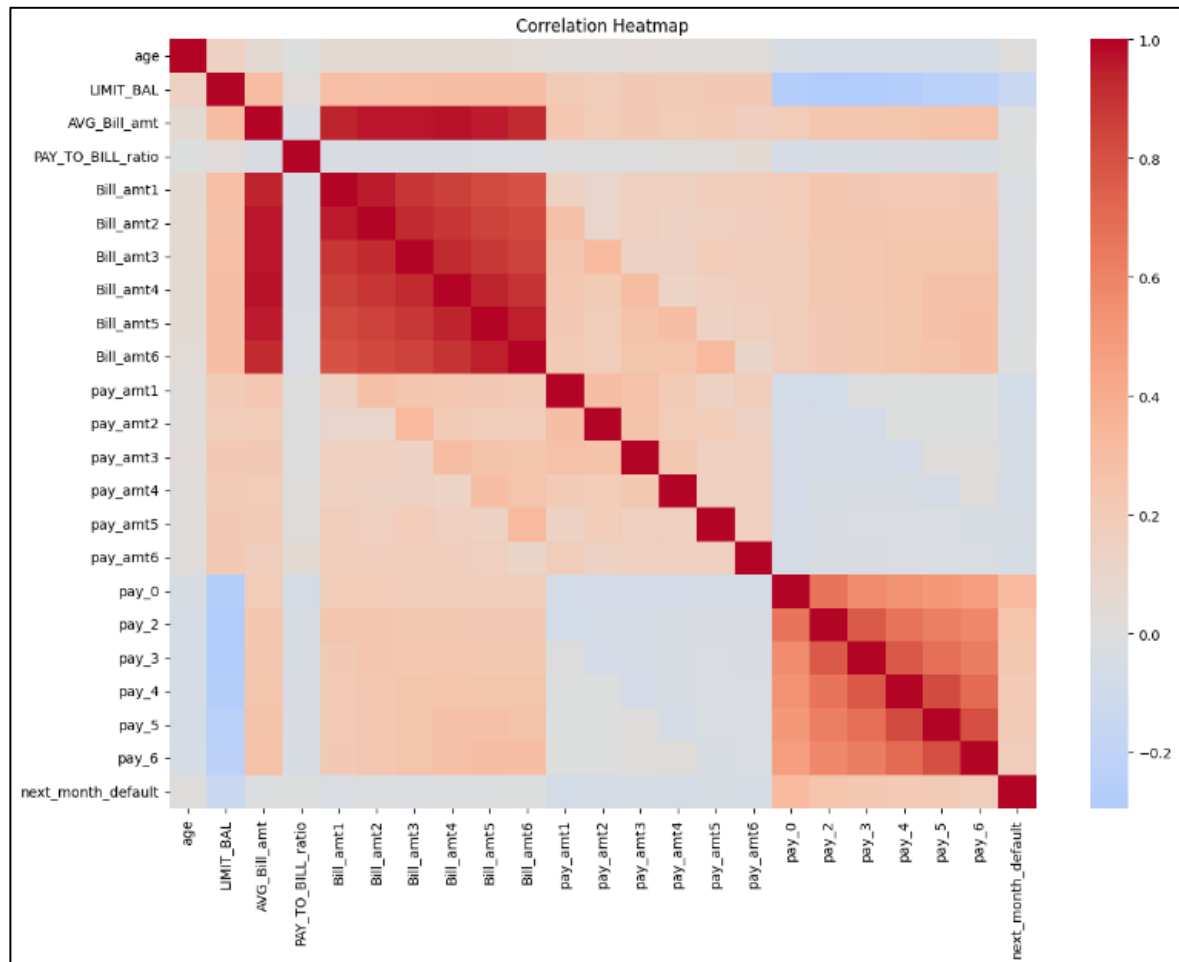
3. Target Variable Analysis

The target variable, **next_month_default**, is imbalanced: the majority of customers do not default, while a smaller proportion do. This imbalance highlights the need for appropriate handling in modeling (e.g., SMOTE, class weighting).

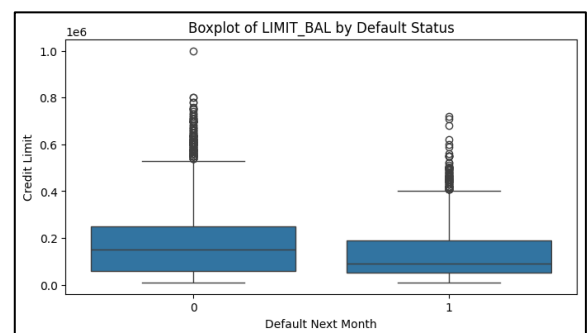


4. Bivariate Analysis and Correlations

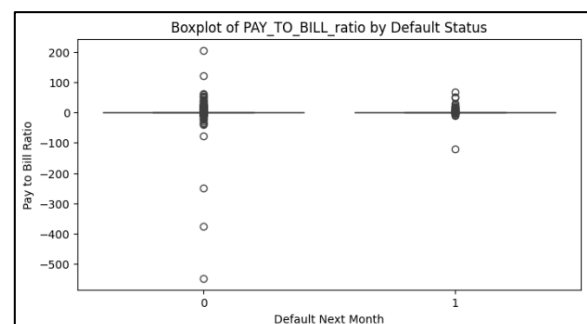
Correlation Matrix: Strong positive correlations were observed among bill amounts across different months and among payment amounts, indicating consistent spending and repayment patterns.



LIMIT_BAL shows moderate positive correlation with bill amounts and payments, suggesting that higher credit limits are associated with larger bills and payments.



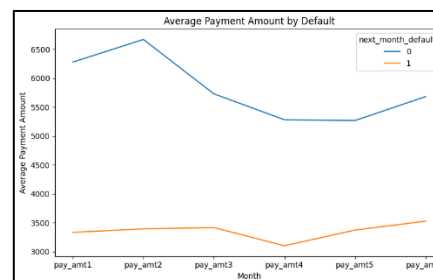
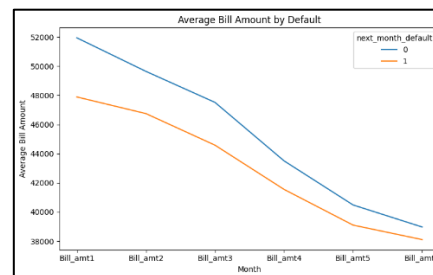
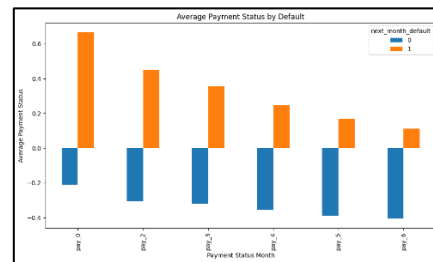
PAY_TO_BILL_ratio is negatively correlated with default, suggesting that customers who pay a higher proportion of their bill are less likely to default.



PAY_0 (most recent repayment status) is positively correlated with default, indicating recent payment delays are a strong risk signal.

5. Behavioral Trends

- Customers with frequent or longer payment delays (higher PAY_m values) have a significantly higher default rate.
- Lower PAY_TO_BILL_ratio and higher average bill amounts are associated with increased default risk.
- Demographic differences (e.g., age, education) show subtle trends but are less predictive than behavioral variables.



Financial Analysis: Key Variables Driving Default Risk

1. Overdue Payments and Repayment History

Repayment Status (PAY_0 to PAY_6):

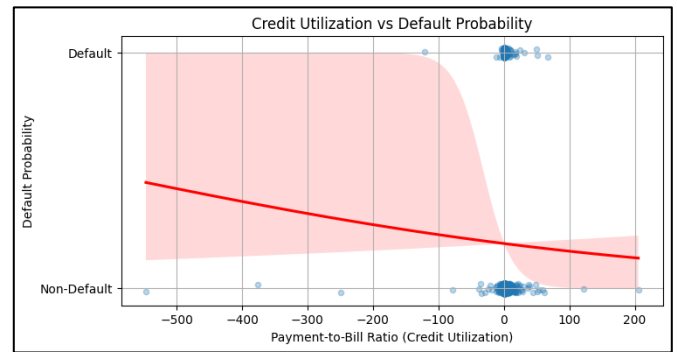
- The countplots for PAY_0 to PAY_6 show that most customers pay on time or make minimum payments (values -1 or 0), while fewer have overdue payments (values 1 or higher).
- This pattern is consistent across all months, highlighting that regular repayment is common, but overdue payments—though less frequent—signal higher default risk.
- Financially, this pattern highlights that while most customers demonstrate good credit discipline, those with frequent or recent overdue statuses are at higher risk of default, making these variables strong predictors for credit risk modeling.

2. Credit Utilization

Credit Utilization Ratio (PAY_TO_BILL_ratio):

- The Payment-to-Bill Ratio measures how much of a customer's total bill is being repaid relative to the total billed amount over a period.

- The plot of this ratio against default probability shows a clear inverse relationship: as the Payment-to-Bill Ratio increases—meaning customers pay a higher share of their bills—the likelihood of defaulting in the next month drops.



- Conversely, customers with low or negative ratios, who pay little or miss payments, are much more likely to default.
- The downward slope of the logistic regression line in the plot visually confirms this trend: higher repayment relative to the bill reduces default risk, while low repayment (or high credit utilization) increases it.

3. Bill Amounts and Payment Patterns

Bill Amounts (Bill_amt1 to Bill_amt6, AVG_Bill_amt):

- The countplots for PAY_0 to PAY_6 show that most customers have repayment status values of -1 or 0, meaning they usually pay their bills on time or make minimum payments. A
- smaller group has values of 1 or higher, indicating overdue payments, but these are much less common.
- This pattern is consistent across all six months, highlighting that while most customers are regular payers, a minority frequently delay payments—an important signal for default risk.

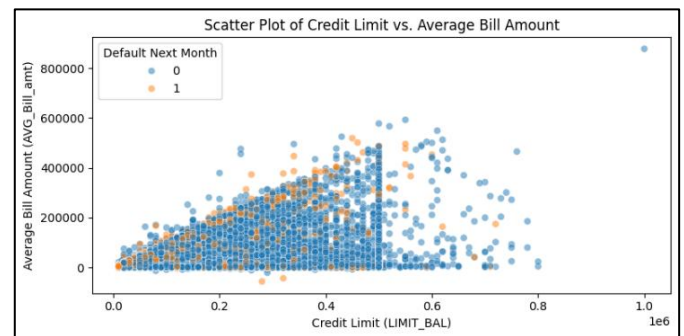
Payment Amounts (pay_amt1 to pay_amt6):

- The distribution plots for pay_amt1 to pay_amt6 clearly show that most customers make small monthly payments, with payment amounts heavily concentrated at lower values and only a few customers making large payments.
- This trend is consistent across all six months, highlighting that substantial repayments are rare outliers.
- Such a skewed pattern suggests that while regular payments are common, high payments are infrequent, which may indicate limited repayment capacity or a tendency to pay just the minimum due.

4. Credit Limit (LIMIT_BAL)

- A higher credit limit gives customers more spending flexibility, but those who use a large portion of their limit (high utilization) are at greater risk of default.
- Similarly, customers with low limits who consistently spend near their maximum may also struggle to repay and default.

- The scatter plot of credit limit versus average bill amount, colored by default status, shows a positive correlation—customers with higher credit limits tend to have higher bill amounts.



- Defaulters are more concentrated among those with moderate to high bill amounts relative to their credit limit, indicating that high credit utilization is closely linked to default risk. Most customers, including defaulters, cluster at lower to moderate credit limits and bill amounts, with only a few outliers at very high values.

5. Demographic and Socioeconomic Factors

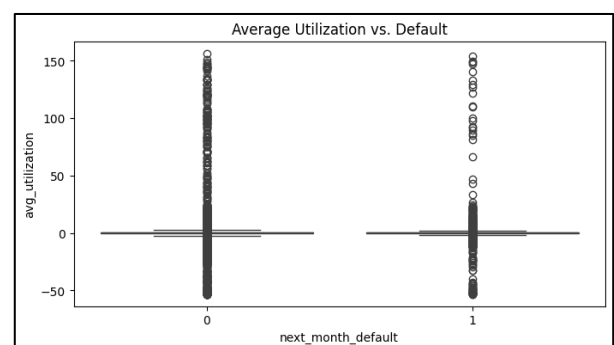
Age, Education, Marital Status, and Gender:

- While behavioral variables are the primary drivers, demographic factors also play a role.
- This analysis shows that most credit card customers are between 25 and 45 years old, with the age distribution slightly right-skewed.
- The dataset is fairly balanced by gender, with similar numbers of male and female customers.
- University or graduate-level education is most common, while the majority of customers are either married or single, with few in the 'others' category.
- These trends indicate a young to middle-aged, well-educated customer base, and no strong imbalance by gender or marital status

Feature Engineering: Creation of New Features

1. Average Credit Utilization Ratio

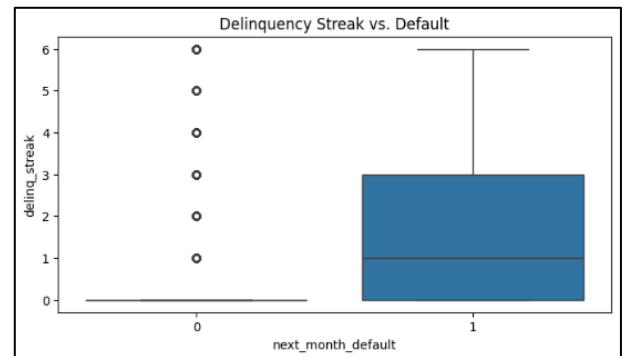
- It is the average proportion of a customer's credit limit that is utilized across the last six months. For each month, bill amounts were divided by the credit limit, and then averaged over six months.
- This boxplot compares the average credit utilization ratio for customers who did not default (0) and those who defaulted (1) in the next month.



- It shows that defaulters tend to have higher average utilization, indicating that customers who use more of their credit limit are more likely to default.

2. Delinquency Streak

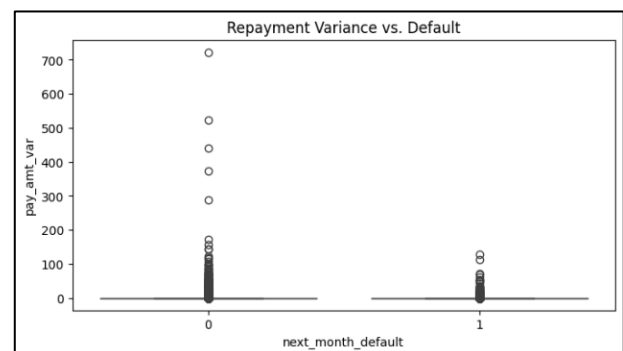
- It is the longest consecutive streak of overdue payments (where repayment status is 1 or higher) in the last six months. Created By scanning monthly repayment status variables (pay_0 to pay_6) and counting the maximum consecutive overdue months.
- This boxplot, "Delinquency Streak vs. Default," compares the maximum number of consecutive overdue months for customers who did not default (0) and those who defaulted (1).



- It shows that defaulters typically have much longer delinquency streaks, while non-defaulters mostly have no overdue streaks. This indicates that persistent overdue payments are strongly linked to a higher risk of default.

3. Repayment Variance

- It is the variance in payment amounts over the past six months. Created by calculating the variance across `pay_amt1` to `pay_amt6`.
- This boxplot shows the relationship between delinquency streak (the longest run of overdue payments) and default status. Customers who defaulted (1) typically have much longer delinquency streaks, while those who did not default (0) mostly have no overdue streaks.



- This indicates that persistent overdue payments are strongly linked to a higher risk of default.

4. Rolling Averages and Trends

- 3-Month Average Bill Amount: Mean of the most recent three months' bill amounts.
- 6-Month Bill Trend: Difference between the most recent and oldest monthly bill amount.
- It matters to capture the short-term spending patterns and emerging risk trends.

5. Log-Transformed Features

- They are log-transformed versions of highly skewed features (e.g., average bill amount, repayment variance, repayment mean, 3-month bill average).
- They matter as they reduce skewness and help models better capture relationships in variables with long-tailed distributions.

6. Interaction Term: Utilization × Delinquency

- It is the product of average utilization and delinquency streak.
- It matters as it captures compounding risk for customers who both over-utilize credit and have frequent overdue payments.

Model Comparison and Justification for Final Selection

A systematic comparison of multiple classification algorithms was conducted to identify the most effective model for predicting credit card default. The models evaluated **included Logistic Regression, Decision Tree, XGBoost and LightGBM**. The performance of each model was assessed using cross-validation and key evaluation metrics such as F1-score, recall, and accuracy, with a particular focus on F1-score and recall due to the business priority of minimizing missed defaulters in an imbalanced dataset.

Model Performance Overview:

- **Decision Tree:** Achieved the highest cross-validated score with a best F1-score of approximately 0.79. The optimal hyperparameters included the 'entropy' criterion and a maximum depth of 5.
- **LightGBM:** Closely followed with a best score of about 0.78, using a learning rate of 0.1, max depth of 5, and 50 estimators.
- **XGBoost:** Also performed well, with a best score near 0.78, and similar hyperparameters to LightGBM.
- **Logistic Regression:** Lagged behind the tree-based models, with a best score of approximately 0.64, despite using class weighting to address class imbalance.

Justification for Final Model Selection:

- The Decision Tree classifier was chosen as the final model because it achieved the **highest F1-score**, balancing precision and recall, which is essential for minimizing missed defaulters in an imbalanced dataset.
- Decision Trees provide strong interpretability, making it easier for stakeholders to understand the main factors behind default predictions—an important requirement in banking risk management.
- Although XGBoost and LightGBM performed similarly well, the Decision Tree's transparency and straightforward explanation made it better suited for deployment and regulatory needs when the performance difference was minimal.
- Logistic Regression was less effective, likely due to its inability to capture the complex, non-linear patterns present in the behavioral and financial features of the data.

Model Comparison Results:			
	model	best_params	
1	Decision Tree	{'criterion': 'entropy', 'max_depth': None, 'm...	
3	LightGBM	{'learning_rate': 0.1, 'max_depth': 5, 'n_esti...	
2	XGBoost	{'learning_rate': 0.1, 'max_depth': 5, 'n_esti...	
0	Logistic Regression	{'C': 1, 'class_weight': 'balanced', 'solver':...	
best_score			
1		0.789727	
3		0.783593	
2		0.783054	
0		0.636109	
DecisionTreeClassifier			
DecisionTreeClassifier(criterion='entropy')			

Evaluation Methodology

The evaluation methodology here prioritizes the **F2-score** and the **ROC AUC score**.

- **F2-score:** This metric is chosen with a beta value of 2, which gives twice as much weight to recall as to precision.
 - **Relevance to Credit Risk:** In credit risk, recall is crucial because it minimizes the number of actual defaulters (True Positives) that the model fails to identify (False Negatives). Missing actual defaulters can lead to significant financial losses for credit institutions. Therefore, by prioritizing recall, the model aims to reduce the risk of granting credit to individuals who are likely to default, making it highly relevant for managing credit risk.
- **ROC AUC score:** The Area Under the Receiver Operating Characteristic Curve measures the overall performance of a binary classifier and its ability to distinguish between the two classes (defaulters and non-defaulters) across various threshold settings.
 - **Relevance to Credit Risk:** A higher AUC score indicates that the model is better at differentiating between those who will default and those who will not. This is important for credit risk assessment as it provides a comprehensive measure of the model's predictive power.

Metrics result

Optimal Threshold: 0.100, Optimal F2-Score: 0.8294				
	precision	recall	f1-score	support
No Default	0.83	0.80	0.82	4088
Default	0.81	0.83	0.82	4088
accuracy			0.82	8176
macro avg	0.82	0.82	0.82	8176
weighted avg	0.82	0.82	0.82	8176

The evaluation results on the training dataset for the credit card project are as follows:

- **Accuracy:** 0.82
- **F1-score:** 0.82
- **Recall:** 0.82
- **F2-score:** 0.82

These identical values across the specified metrics (accuracy, F1-score, recall, and F2-score) for the training set suggest that the model achieved a consistent level of performance during the training phase.

Classification Cutoff

Selected the classification cutoff, also known as the optimal threshold, by analyzing the **F2-score** across various possible thresholds on the validation dataset.

Here's a detailed breakdown of the process:

1. **Probability Prediction:** The trained model generates probability scores for each instance in the validation dataset, indicating the likelihood of that instance belonging to the positive class (e.g., defaulting).
2. **Precision-Recall Curve Analysis:** The `precision_recall_curve` function is used, which calculates precision and recall values for a range of probability thresholds.
3. **F2-Score Calculation:** For each threshold, the F2-score is calculated. As previously explained, the F2-score places a higher emphasis on recall, making it a suitable metric for credit risk where identifying actual defaulters is crucial.
4. **Optimal Threshold Selection:** The threshold that yields the maximum F2-score is then chosen as the `optimal_threshold`. This threshold is subsequently used to convert the predicted probabilities into final binary classifications (default or non-default).

By maximizing the F2-score, it ensures that the chosen cutoff effectively balances the trade-off between precision and recall, with a deliberate bias towards maximizing the identification of true positive cases (defaulters) in the context of credit risk.

Business implications

The credit risk model primarily helps financial institutions by:

- **Minimizing Financial Losses:** Accurately identifying potential defaulters reduces bad loans.
- **Improving Risk Management:** Enables data-driven, informed lending decisions.
- **Optimizing Loan Portfolio Quality:** Leads to a healthier and more stable portfolio.
- **Boosting Operational Efficiency:** Facilitates faster and potentially automated credit assessment.
- **Strategic Resource Allocation:** Helps focus resources on high-risk cases while streamlining low-risk ones.

In short, the model shifts credit management from reactive to proactive, ensuring more stable growth and profitability.

Summary & Key Findings

Summary of Findings

1. **Prioritized Evaluation Metrics:** The project highlights the **F2-score** and **ROC AUC score** as the primary metrics for evaluating the credit risk model. The F2-score (with a beta value of 2) was specifically chosen to emphasize **recall**, reflecting a strategy to effectively identify potential defaulters.
2. **Training Dataset Performance:** The model demonstrated consistent performance on the training dataset, achieving **0.82 for Accuracy, F1-score, Recall, and F2-score**. This indicates the model learned the underlying patterns in the training data quite well.
3. **Classification Cutoff Selection:** The methodology for selecting the classification cutoff (optimal threshold) involved determining the point that **maximized the F2-score on the validation dataset**. This method reinforces the project's focus on optimizing the model's ability to identify true positive cases.

Key Learnings

1. **Strategic Metric Selection:** A key learning is the importance of selecting evaluation metrics that align precisely with the problem's specific needs. Prioritizing recall through the F2-score in a credit risk context is a direct application of this principle.
2. **Threshold Optimization:** The project demonstrates a practical approach to optimizing the classification threshold using the F2-score on a validation set, which is crucial for translating model probabilities into actionable binary decisions.
3. **Model Performance Insight:** The consistent performance across various metrics on the training set provides initial confidence in the model's learning capabilities, although further evaluation on unseen data (e.g., test set, if available) would be necessary for a complete assessment of generalization.